

Health & Safety

Standard Emergency Management



Reference

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1. Objective

The objectives of this document are to:

- Describe the management methods to be implemented on the project
- Establish the roles and responsibilities of the main parties involved
- Ensure all emergency levels are properly managed and reported
- Define communication and authority flows when managing emergencies

This document does not address emergency situations related to security, financial, legal or administrative issues.

The specifics of underground work emergencies are addressed in annex **Emergency measures in underground environments [BYTP-INF-H&S-2172]**, which is associated with Reference **Standard Underground work activities [BYTP-REF-H&S-2227]**.

The specificities of emergencies in a dry atmosphere hyperbaric environment (tunneling) are addressed in the *BYTP hyperbaric work safety manual mention D* of BYTP.

The management of environmental emergency situations is defined by the BYTP Environment Quality department through the procedure ENV-PRO-3005-B-EMERGENCY SITUATIONS

2. Scope

This document applies to all operations carried out on all BYTP projects/sites and in the event of an emergency situation such as:

- Work accident
- Road accident
- Fire
- Environmental pollution
- Instability of a construction
- Natural disaster
- Industrial accident near our construction sites

This document applies to all companies working on sites (including service providers and subcontractors).

In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

3. Documents reference

- N / A

4. Definitions and abbreviations

Term / Abbreviation	Description
Emergency situation	A sudden event which presents an immediate risk of serious harm to the health and safety of people, equipment, property and/or the environment. Emergencies can result from human behavior, malfunctions/failures of materials, equipment or installations, events of a technical nature (short circuit, explosions, fire, etc.) or natural events (earthquakes, floods, etc.). It is an event which requires immediate action, the direct scope of which is generally limited to a perimeter.
Emergency scenario	An event that is likely to generate an emergency situation. Emergency scenarios are tools for forecasting possible events that may occur during the execution of the project and suffered by the personnel (and assets) of BYTP and all stakeholders involved. They are defined, among other things, on the basis of historical data of events occurring at BYTP (and more generally in the Public Works sector) and whose occurrence is considered probable.
Central Emergency Team (CMT)	Temporary body at the service of the Delegate of Power / Project Manager to help him manage an emergency situation on his site / project
LIC	Local Incident Correspondent
PIC	Principal Incident Correspondent

5. Roles and Responsibilities

Depending on the typology and size of the projects, sites or entities, the roles and responsibilities, described for each actor in the management of emergency situations, are assigned by name in the Emergency Plan of the project/site.

On large projects/sites with organizations allowing it, dedicated people are designated to carry out these missions.

For projects/sites with a smaller organization in terms of management, roles can be assigned as follows:

- PIC : to the delegated authority or one of his deputies or to the Site Manager
- LIC : to the Works Manager or the Site Manager

In the context of non-production sites (agency, offices, laboratories), the nominative allocation of roles is specified in the site Emergency Plan.

5.1. Project Manager

- Guarantees the implementation of this document on its project
- Mobilises the necessary resources
- Ensures personnel are properly selected and trained in emergency and response procedures
- Assesses the severity of the emergency
- Designates the Principal Incident Correspondent (PIC) or may act as such
- Designates the Local Incident Correspondent (LIC)

5.2. Principal Incident Correspondent – PIC

- Coordinates level 2 and 3 emergency interventions (See chapter 8)
- Is responsible for managing emergency situations
- Activates the Central Emergency Team (CET)
- Must be able to arrive on site as quickly as possible and, in any case, in less than an hour
- Ensures the interface between the site/project and the command of external intervention services
- Has all the elements related to the work in progress
- Is able to take all necessary measures for the proper running of the emergency responses
- Notifies its operational entity and keeps it regularly informed of developments in the situation
- Ensures effective communication with authorities, the media and the public

5.3. Local Incident Correspondent - LIC

The LIC, designated by the Project Manager, is a trained member of the works management, having the material and technical knowledge necessary for emergency management on site. One LIC is permanently present on site during the working hours and:

- Ensures logistics related to emergency management
- Supervises a team whose role is to ensure the guidance of rescue operations and logistical support for rescue personnel (lifting, transport of personnel and/or equipment, etc.)
- Is the on-site contact for external intervention services
- Allows on-site access to external intervention services
- Informs the emergency services of the situation of dangerous networks and products
- To any authority over all personnel present, whatever the hierarchical level and/or the company they belong to (within the framework of their emergency management mission)

5.4. Works manager / site manager (superintendent)

- Makes its teams available to the LIC in the event of an emergency situation
- Is associated with the management of level 1 emergencies occurring in its activities (See chapter 8)
- Coordinates the activities of its personnel involved in the inspection, maintenance and use of emergency equipment
- Itself available to the PIC upon activation of the Central Emergency Team (CGU)

5.5. Project Methods Manager

The Project/Site Methods Manager establishes and maintains the Site Installation Plan by indicating:

- Access and circulation including any specific “Emergency” access
- The means available to respond to an emergency situation (medical service, assembly points, fire extinguishers, tunnel emergency room, safety showers, fire hoses, access, circulation, etc.)
- Materials likely to have an impact on the aggravation of an emergency situation (gas storage, storage of combustible products, storage of dangerous products, LV/HV electrical lines, gas pipes, pipes of combustible products, circuits under LP/HP pressure, etc.)
- Equipment that can be made available or can facilitate the missions of public emergency services (lifting equipment, transport equipment, emergency manbasket, MEWP, etc.)
- Any changes related to the above

The Site Installation Plan is made available to the emergency services upon their arrival on site and is shared with them as many times as necessary.

5.6. Health and Safety Manager

- Assists the Project Manager in the implementation of this document on the project/site
- Drafts and maintains the Project Emergency Plan
- Integrates key elements related to emergency management into the induction module
- Organizes the training of project teams in emergency intervention
- Informs the Project Manager of any emergency situation
- Ensures that all emergency equipment (detection and response) is available and maintained in good working order at all times

In the absence of a H&S Manager, all of the above responsibilities are assigned to a person formally designated by the Project Manager.

5.7. External Rescue Services

External Rescue Services are an integral part of effective management of emergency situations, when their intervention proves necessary.

Their intervention remains under their entire responsibility.

However, from the start of the work, the Project must invite the various emergency services to the site in order to:

- To present to them the project and the issues in terms of managing the different emergency situations that may occur
- To communicate the Site Installation Plan to them (at each revision)
- That they become aware in situ of the conditions of access to the site(s), according to the different phases of activities
- That their intervention procedures are in line with those of the Project

When the Project/site is of unusual dimensions and/or of an unusual design, regular visits are carried out with emergency service personnel so that they have an overall vision of the Project/sites (construction of an underground station, viaduct, intervention in a tunnel, etc.)

The terminologies and acronyms used by the Project and by the Emergency Services may be different. It is then necessary to define a common language in shared documents.

5.8. Any other participant

All personnel working on the site/project (including visitors):

- Takes note of and applies the project's emergency instructions
- Informs his manager if he witnesses an emergency situation (of any kind)
- Intervenes, within the framework of its capacities, skills and means, in any dangerous situation which it witnesses

6. Training and competencies

6.1. On-site induction

During their induction on site/project, all personnel are informed of the applicable safety and emergency instructions and the management procedures to follow.

6.2. Training

The Project/site identifies the people capable of managing emergency situations at any time. These personnel are trained and in sufficient numbers to compensate for potential absences (vacation, illness, etc.).

A training program is defined and implemented by the project/site for the different actors involved in emergency management. This training can be delivered internally or through external training organizations and covers, among other things:

- Regulatory training at the workplace including emergency management at the workplace, securing equipment, guiding emergency services, etc.
- First aid training
- First aid training in an underground environment (if applicable)
- Training in handling fire extinguishers
- Evacuation training

Any safety training not mentioned above is used in the context of managing emergency situations (asbestos training, working at height, diving, etc.).

7. Materials and equipment

Each site/project defines and puts in place the necessary equipment to effectively carry out emergency actions.

This equipment can be specifically dedicated to emergencies or common equipment.

See document **Emergency measures in underground environments [BYTP-INF-H&S-2172]** and the BYTP *hyperbaric work safety manual mention D* , for specific emergency equipment associated with these activities.

7.1. Rescue guidance

At a minimum, emergency guidance equipment consists of:

- Autonomous magnetic flashing lights
- Traffic cones
- Tape rolls
- Expandable linkable lightweight barriers
- Rechargeable LED sticks (with charger)
- Luminescent Survival Sticks

The quantity of material is defined at the start of the Project and is validated during full-scale drill.

7.2. Lifting

The Project/site may, as part of an emergency response, be required to carry out lifting of equipment and/or personnel.

Different types of lifting should be considered in the context of rescue operations and are anticipated/managed in accordance with the **Lifting standard [BYTP-REF-H&S-2210]** :

- Common liftings
- Specific liftings (emergency man-basket for example)

7.3. Medical emergency

Project/site first aiders may also be required to intervene. The Project/site must have at least the following lifesaving appliances:

- First aid kits
- First aid kit reconditioning lots
- AED (Automatic External Defibrillator)
- Craneable stretcher
- etc.

8. General emergency management

Managing an emergency situation consists of activating, immediately and without delay, actions whose sole purpose is to prevent any serious harm to people and/or property.

An emergency is managed differently depending on its level (see table below).

Emergency level	Description	Management manager emergency _	Persons/Entity(s) who need to be informed in order to advise in case of worsening emergency
Level 1	Minor event that can be managed at the site/project level with staff and equipment available on site. This situation does not require the intervention of external emergency services. <i>Examples of level 1 emergency:</i> ▪ Minor injury with on-site treatment ▪ Start of fire easily controlled ▪ Leak of dangerous products controlled (portable means)	Site Manager (superintendent)	Production Manager
Level 2	Significant event requiring the intervention of project/site resources and equipment. This incident is controlled by site/project resources. Local authorities and public administrations are informed and may be required to intervene to close the incident. The customer is informed. <i>Examples of level 2 emergency:</i> ▪ Injury with transfer to local healthcare facility for examination ▪ Fire dealt with by internal means. Local firefighters validate the extinction ▪ Significant leak of dangerous products in internal area and treatment by internal means	LIC	▪ PIC ▪ Production Manager ▪ Project Director
Level 3	Serious event requiring the intervention of resources and equipment beyond the scope of the project/construction site with the help of local authorities and public administrations The customer is informed. The Central Emergency Team can be activated. <i>Examples of level 3 emergency:</i> ▪ Injury treated by external services with transfer to local care structure ▪ Fire dealt with by local firefighters ▪ Significant leak of dangerous products in internal area and treatment by external means	LIC Or PIC (as soon as the CET is activated)	▪ PIC ▪ Project Director ▪ Operational management Dept. (UO/country/headquarters) ▪ H&S Management Dept. (UO/country/headquarters)
Level 4 Crisis	Major event whose scale and/or consequences extend beyond the scope of the project and which requires: ▪ Help from local authorities and public administrations ▪ Activation of the Central Emergency Team The customer is informed. <i>Examples of Crisis situations:</i> ▪ Death ▪ Multiple victims and/or victims whose vital prognosis may be in jeopardy. ▪ Major fire with intervention by local firefighters ▪ Significant leak of dangerous products in external area and treatment by external means ▪ Impact on a third party (collapse, drop of load on public roads, etc.)	PIC	▪ Project Director ▪ Operational management Dept. (UO/country/headquarters) ▪ H&S Management Dept. (UO/country/headquarters) ▪ BYCN security Dept. for International projects

The operational and H&S Departments of the UO/country/headquarters are notified and remain available to the Project/site in the management of the emergency in accordance with the procedure **Management of Health & Safety events [BYTP-PRO-H&S-2082]** .

8.1. Intervention of external emergency services

All types of emergency situations (except level 1) require coordination of resources with external emergency services.

These services may, depending on the situation, need to be present on site.

When an emergency situation is triggered, the number of vehicles and/or external emergency equipment traveling to the site can be very large. Their parking locations and positions are coordinated between the site/project and external stakeholders (police, ambulances, firefighters, etc.).

This point is essential, otherwise it will slow down the progress of public emergency services towards the on-site intervention site and consequently help people will be delayed.

3 parking levels are generally required:

- Level 1: vehicles in intervention → priority parking areas
- Level 2: reinforcement vehicles in → nearby parking areas
- Level 3: vehicles in connection (no need for intervention) → remote parking areas

The Project/site identifies parking areas according to the levels above and makes plans available to the guard post(s) and staff of the team in charge of receiving emergency services. Each emergency vehicle driver must know their parking priority (1, 2 or 3) depending on their role in the emergency response.

8.2. Central Emergency Team (CET)

A Central Emergency Team (CET) is a temporary structure, set up by the Project Manager, to enable an organized and efficient handling major event.

Its objectives are to:

- Collect information
- Structure the information
- Analyze the information
- Enable decision-making
- Disseminate information
- Disseminate decisions

It is triggered by the Project Manager:

- Possibly in the case of a level 3 emergency
- Systematically in the case of a Crisis situation (level 4)

The Central Emergency Team is sized according to the event and its consequences, both internal to the project and externally (attacks on people and public property, etc.).

8.3. Members of the Central Emergency Team

The main members of the Central Emergency Team (CET) are:

- A manager (usually the PIC)
- A coordinator
- A secretary (keeping logbook, telephone, etc.)
- Technical specialists (mobilized on demand)
- A communications team for the authorities and the public
- The head of External Relief Operations and his deputies

Each member of the Central Emergency Team (CET) has one or more back-ups trained and able to fulfill the assigned role.

Depending on the organization of the Project/site, some roles may be merged within the CET.

Annex **Central Emergency Team [BYTP-INF-H&S-2070]** details the roles of the members of the CET.

8.4. Material resources of the CET

The equipment of the Central Emergency Team is dedicated to it and must under no circumstances be used for other purposes, in order to guarantee the technical support necessary for members of the CET to carry out their missions.

The Central Emergency Team must be able to be set up in a sufficiently large room. It is sized according to the number of members of the CET. These dimensions are reassessed/optimized following the various emergency exercises and/or the ramp-up of the project.

Annex **Central Emergency Team [BYTP-INF-H&S-2070]** lists the means available to the CET, so that its operation is optimal.

8.5. Project/site emergency plan

Each Project/site:

- Develops and maintains its specific Emergency Plan which details the preparation and response measures for the various anticipated emergency situations
- Develops and maintains an emergency management flowchart that summarizes the elements of the Emergency Plan (scenario, process, actors, key contacts)
- Develops an annual emergency exercise plan and carries out the exercises in accordance with this plan
- Provide the following elements (and their successive revisions) to interested parties of the project, as well as to the H&S Country/Headquarter Dept. :
 - Emergency plan
 - Emergency management flowchart
 - Planning of emergency exercises
 - Emergency exercise reports

The Project/Site Emergency Plan takes into consideration all contractual requirements relating to emergency management (level 1 to Crisis), such as:

- The emergency management system of the Customer and partners

- The requirements of the Client and partners in the relationship with local authorities and media
- Local and legal requirements

The main objectives of the project/construction emergency plan are:

- To ensure the effective and efficient management of identified emergency scenarios
- Define roles and responsibilities
- To assess the situation and put in place appropriate prevention and control measures
- To define the type and frequency of emergency exercises taking into account the possibility of involving local authorities and potentially affected communities

The Emergency Plan must be:

- Specific to the scenarios identified on the project/site concerned
- Kept constantly up to date
- Accessible and available to all project stakeholders/ external services/ UO/ country/ headquarters

Its development is based on the evaluation:

- Hazards associated with work activities
- Natural, geographic and environmental hazards associated with workplaces
- Hazards linked to the health of personnel involved

The Emergency Plan is reviewed:

- Immediately after each exercise and following any emergency situation actually experienced, in order to integrate any improvement deemed necessary
- When a new emergency scenario is identified
- Following changes that impact the emergency plan in place, such as a new configuration of the site, a change in equipment and materials used, a change in the organization and key personnel

Additional documents may be developed by the project/worksites according to local regulatory requirements or applicable international standards.

For international projects, the Emergency Plan is established in consultation with the BYCN Security Department, in particular for projects where it is necessary to define emergency means managed directly or indirectly (subcontracting) by the project, due to the absence or unreliability of the means locally provided by the authorities.

8.6. Management of internal communications

An effective line of communication is established to obtain the latest information and updates between the parties involved, planned corrective actions and/or mitigation actions and any relevant information to help judge the development of the event and of its possible worsening towards higher levels.

A telephone directory is set up and kept up to date with the holders and their partners.

All actors involved in the emergency management process, on Project/ site/ UO/ country/ headquarters, ensure continuous availability.

8.7. Involvement of authorities

Specific instructions are defined at the Project/site level in order to manage, in the event of an emergency, the arrival/involvement of representatives of organizations/authorities with judicial/criminal power or with investigative power.

In the event of an emergency situation, the Entity/Project provides specific support to develop reports and documentation that may be required by any external governmental body (local, national, international authorities).

8.8. Management of external communications

In the event of an emergency and based on the information received, an appropriate external communications plan is prepared and implemented.

This plan defines, among other things, the list of contacts and their contact details based on plausible emergency scenarios.

In the event of a Crisis, BYTP's Communications and Legal managers evaluate the way in which information concerning the occurrence of the event must be disclosed outside the company, in agreement with BYTP General Management.

In the event that a clause in the contract established with the Client concerns the management of external communication in the event of an emergency, BYTP will provide all the support and information necessary for its implementation.

9. Emergency exercises

9.1. Objectives

In order to test and improve the effectiveness of emergency procedures, periodic exercises are scheduled and are carried out taking into account all identified emergency scenarios. The exercise should make it possible to check the lines of communication, the reaction capacity of personnel and the availability of internal/external equipment and resources involved in emergency management.

The results of the exercises make it possible to assess the competence of intervention personnel, the adequacy of emergency procedures and to define and implement improvement actions, if necessary.

The success of managing a large-scale exercise depends, but not only, on its preparation and in the training of the personnel involved.

9.2. Realisation

Exercises are identified and scheduled in an annual emergency exercise plan taking into account the most stringent requirements of BYTP, the client and national and international regulations.

In addition, unannounced exercises can be organized based on a scenario and a schedule not communicated to the personnel.

An emergency management scenario for working at heights is planned at least once a year.

The exercises are as realistic as possible and simulate foreseeable emergency scenarios at the relevant site/project. They are conducted at different times of the day (including night) to avoid routine and to simulate all possible conditions. These exercises are carried out in such a way as to limit the disruption of current activities and not to generate risks for employees and any interested parties.

It is appropriate to increase the difficulty of the exercises in accordance with the development of the teams' abilities to react and intervene.

The type and frequency of emergency drills conducted in project office areas depends on local legislation, but it is recommended that they be conducted at least once a year.

9.3. Emergency exercise report

Any exercise is subject to a general debriefing with those responsible for the activities concerned and/or impacted and is recorded in an experience report drawn up using the **Emergency drill report [BYTP-FOR-H&S-2171]**.

This report is completed by the H&S Manager and is mainly based on the general debriefing and the observers' reports.

It contains areas for improvement and/or actions to be carried out, at the level of:

- Training
- Organization
- Display
- Material
- Etc.

The actions taken and the improvements to the emergency plans are brought to the attention of all staff through briefings and/or specific training.

10. List of annexes

- Annex 1 – Central Emergency Team (BYTP-INF-H&S-2070)
- Annex 2 – Emergency drill report (BYTP-FOR-H&S-2071)